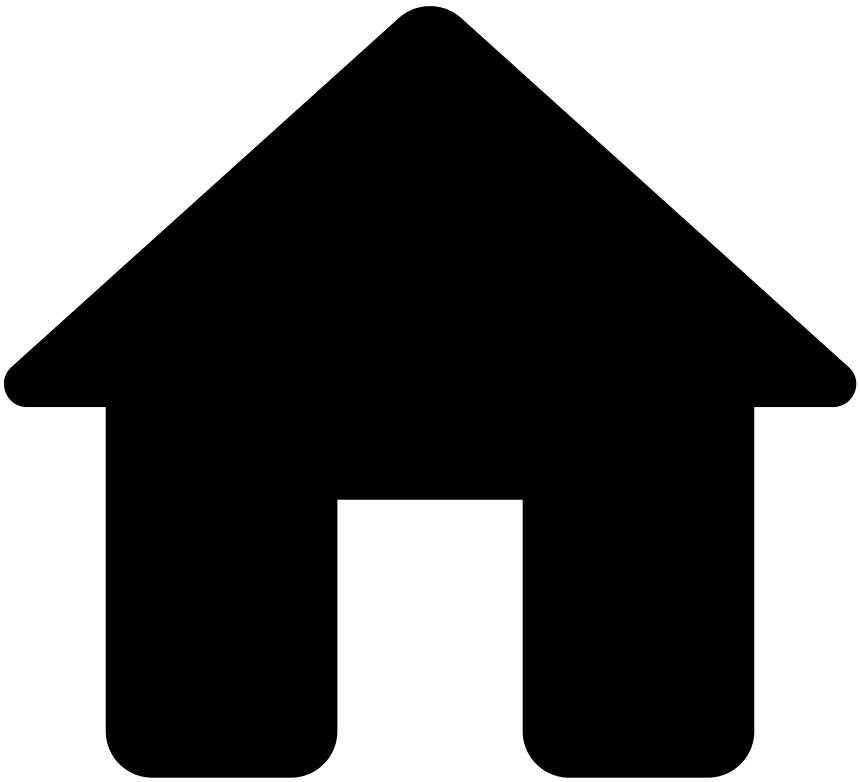


Authentication

Feedzai allows you to authenticate and protect your data through JSON Web Tokens (JWTs). A JSON Web Token (JWT) is a piece of information that is transmitted as a JSON object from clients to Feedzai, and its purpose is to ensure that client requests (e.g. transactions for processing) have not been compromised.

These tokens are generated using the clientâ□□s private key, which can be verified with the corresponding public key. The public key must be shared with Feedzai beforehand to verify that the token sent with the request belongs to the client.

Overviewâ□□

JWTs protect against different attacks using the following mechanisms:

- **Encryption/signing** so that only authenticated users can access sensitive data.
- **Expiration dates** to protect against replay attacks.
- **JWT revocation** to block exposed JWTs as soon as possible.

You can see more details on integrating JWTs with your software in the sub-sections below, but in general you will:

1. Generate a private-public key pair to sign the JWT.
2. Provide the public key to Feedzai in JSON Web Key Set format.
3. Frequently generate new JWTs with a reduced validity period (e.g. 5 minutes). Each JWT is signed with your private key generated in step 1.
4. For each request to Feedzai, include a JWT in the `Authorization` header.

Given the recommended algorithm and parameters, you can choose the tool that best fits your system. A comprehensive list of tools is available at the libraries section of jwt.io.

danger

Third-party security vulnerabilities: When choosing a library, make sure it has no known security vulnerabilities.

JSON Web Keyâ□□

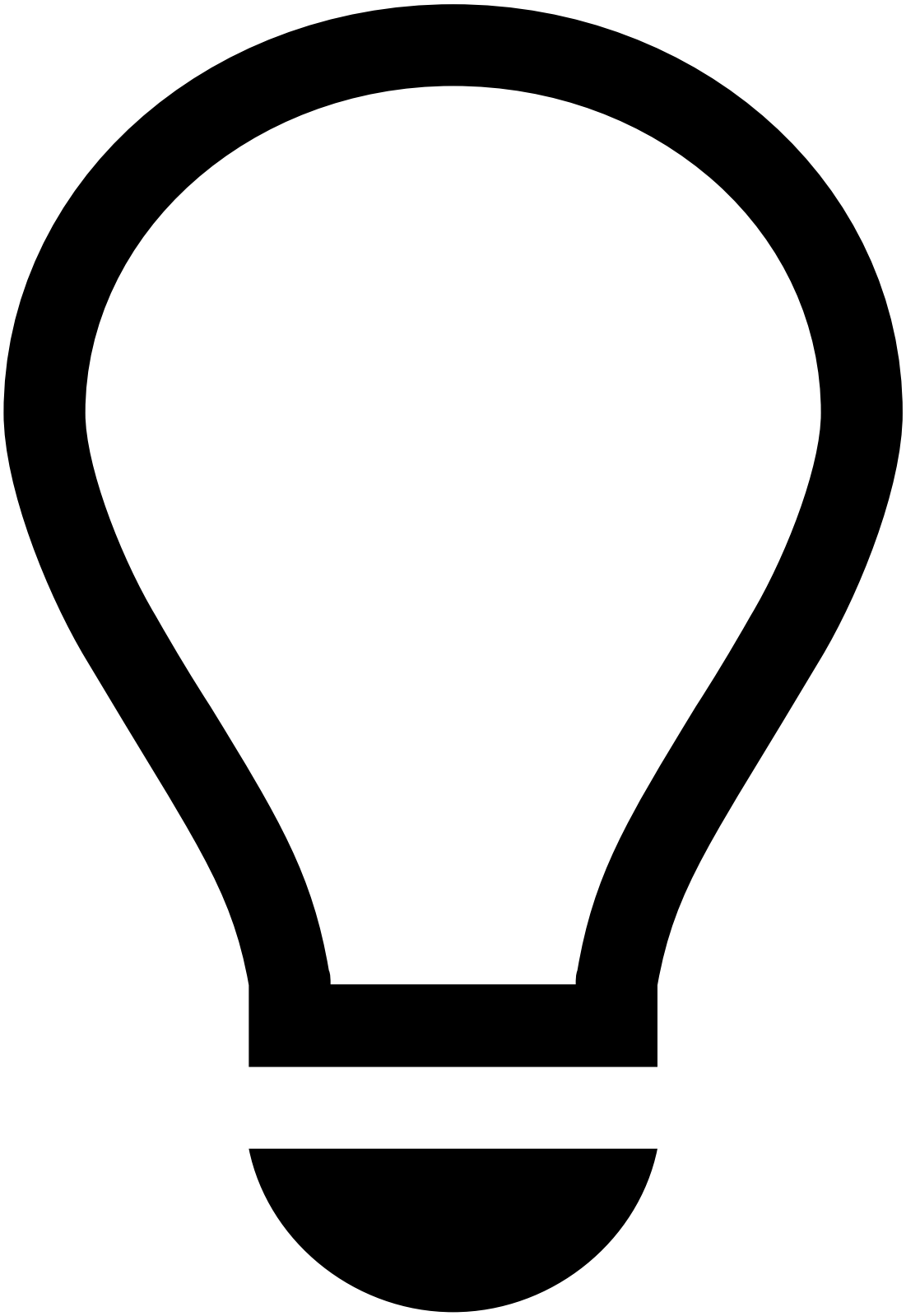
A key pair is necessary to cryptographically sign and verify the JWTs sent with your requests.

- You will use the private key to sign your JWTs.
- Feedzai will use the public key to verify the signature, and therefore the authenticity of the requests.

To create a JWK set, you need to choose the signing algorithm and the JWK parameters.

The following algorithms are supported:

- `RS512` (default)
- `RS256`



tip

Note that each algorithm must be specified in both the JWK and JWT json files - e.g. if you send a `RS256` JWT, you must specify `"alg":"RS256"` in the JWK.

Feedzai recommends the `RS512` algorithm (âRSASSA-PKCS1-v1_5 using SHA-512â) with a 4096-bit key. See the sections below to learn how to generate a JWK set with standard tools.

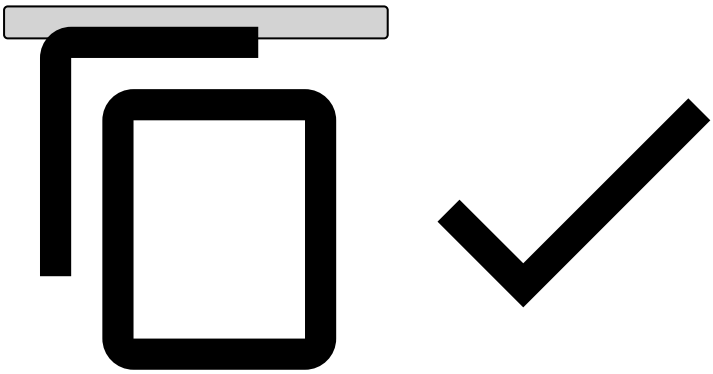
danger

Beware of private key parameters: Construct the JWK set so that you **do not** send any of the private key parameters.

Your JWK set should look similar to this:

```
{
    "keys": [
        {
```

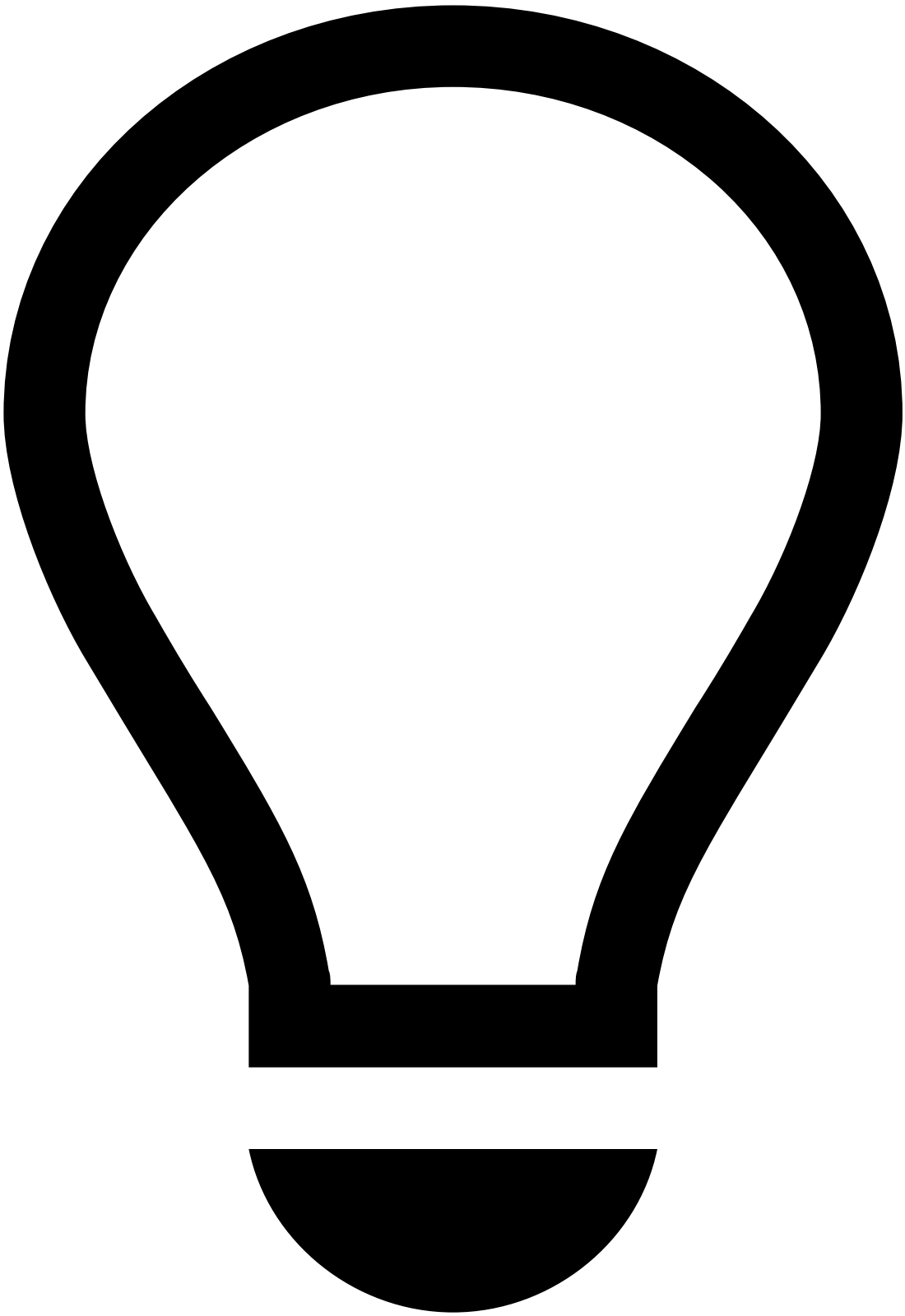
```
            "e": "AQAB",
            "n": "187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqy
Owf6CBv6xj9ZnqyyScle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGB
dGqbvm374XNfLRb0sF7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqRO
HQNyzchbOWgEPUKoRBRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
            "kty": "RSA",
            "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
            "use": "sig"
        }
    ]
}

{
        "kid":"mykey2",
        "kty":"RSA",
        "alg":"RS512",
        "use":"sig",
        "e":"AQAB",
        "n":"k51ZPyu9gocn508aXzob_hSKaIAnJApY4pY3h1oym1Z15-6DN3WxB3YHUyXPRd-iBzF7177XjQAywTzAaz-
PKLZ4bC52noanLx7Ihyf3nQAOwhEv16zaJLI-
HqMEMp6JKPjKJ8LKanVnvxWNI0zUBwcBOwt9cxph0evyVO94cXA0RZxUZxa7C3FuSYkO2zWorMH0ZkC3harkJx0C9Q7nBiGW
PQmB8ZkLx04iWlm8ldyxDJPKl-PC4-
gS3y2mgPfZVAaK_VGod88PUmF4vFax7a03v2VLtn9wm8UWs9SXATxJ5cExgIIvFkdsZAcS3hWnhUe3Of4zAEbKh2Ah5DtfY9
xvtsU8FGlak7wLCwVnubWWglI2hmFvFU7Jsyzb_L-
jD7XCYrg7QuFV8mX2jVcTbsSxnd8GLRNbvmztvsjfaQHXEHraTRJFoGO1Y3JXDiY8Kj40OQXxBNXaFxHda9lBfzWfDkgWvcy
eiTPeWpKc6cS1zcDaQNRzZr3aZR-
gI4U8YTAXcIKDdom2GCQSx9WfmFq3uAj95duL5CgyGfw7L-02c6ckRr6kBAKebURkSsZXOyTO8bdbS8GzPMD59CrRL2SnJJe
qHyrEav8JCu0x3UCXbyqlfqQgzVsXNpFUMIS-sUhe3nyVQOeJ_WoYq7oAbjLpx_Bw5elQJf4zJbJd2Pk"
    }
```



In the aforementioned example, the following parameters are used:

Name	Extended name	Description
`e`	Exponent	RSA public exponent e.
`n`	Modulus	RSA public modulus n.
`kty`	Key Type	Should be `RSA` to symbolize you are using the RS512 algorithm.
`kid`	Key Id	Should specify a key identifier of your choice.
`use`	Public Key Use	Should be `sig` to note this key will be used to verify JWTs.

How to generate a JWK Set (JWKS)â



tip

This section describes how to generate a JWKS using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

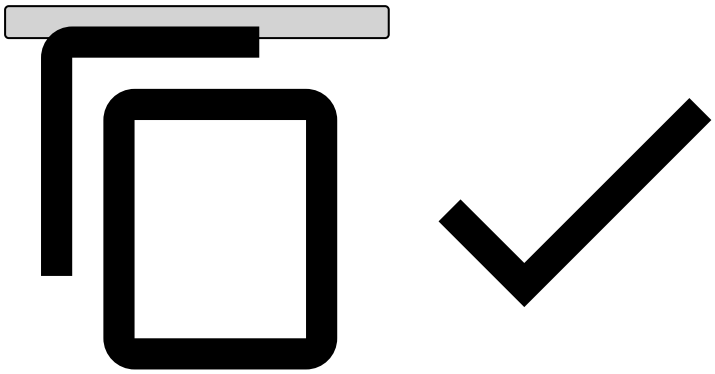
Consider the following requirements if you're generating JWKSs in a different way:

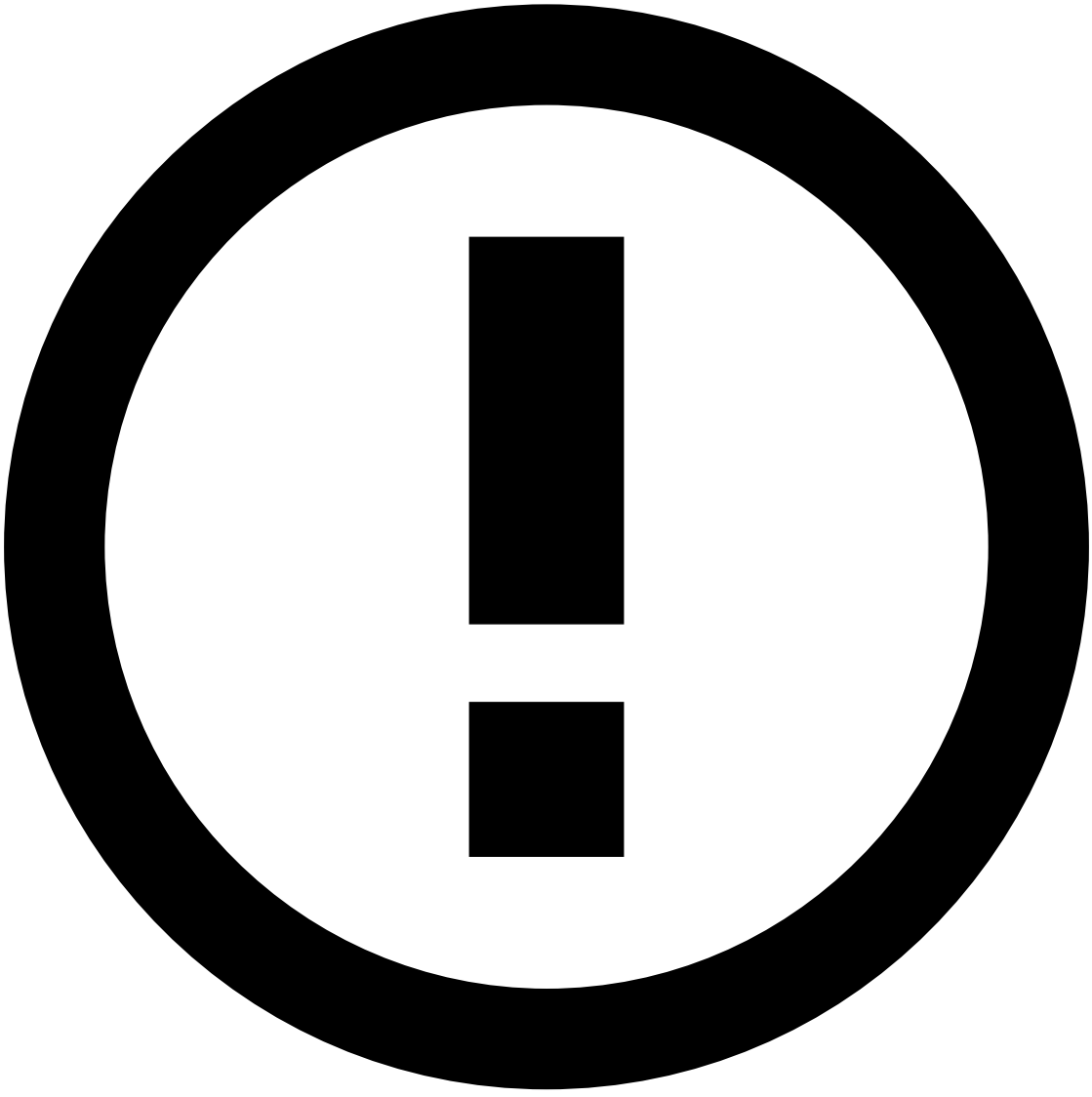
- Make sure you store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**

- Manage JWKs in Pulse.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```



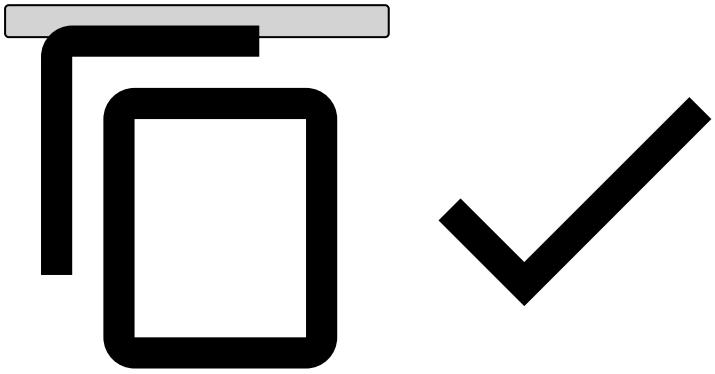


info

Make sure you repeat steps 1 through 3 to create a public-private key pair for each environment.

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
 --mode JWK \
 --generateKeys \
 --privateKey "/path/to/privateKey" \
 --publicKey "/path/to/publicKey" \
 --kid "key_id"
```



The JWK Set JSON will be displayed on your standard output. Alternatively, you can use the `--jwk` parameter to specify a file where it should be stored instead.

5. Store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**
6. Manage JWKs in Pulse.

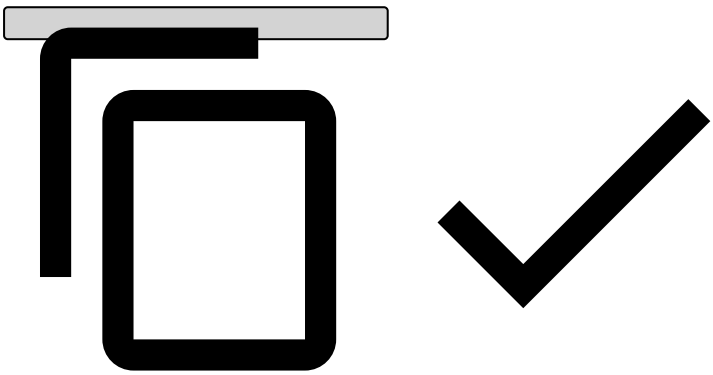
JSON Web Tokensâ□□

Once you have a private key, you can generate your JWTs. All JWTs are composed of a set of headers, a set of claims and a signature.

JWT Headersâ□□

Besides the algorithm (`alg`), the only mandatory header for JWTs is the `kid` (the key ID), which indicates which key was used to sign the token and **must** match a valid key in the JWK Set used for validation. The set of headers should be similar to:

```
{
    "kid": "keyID",
    "alg": "RS512"
}
```



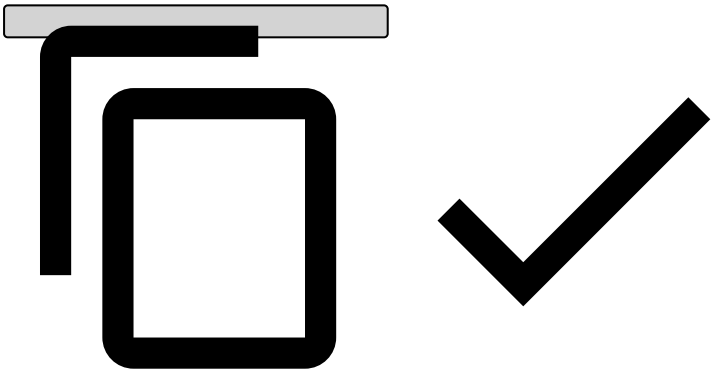
JWT Claimsâ□□

Typically, the following claims should be present on the JWTs:

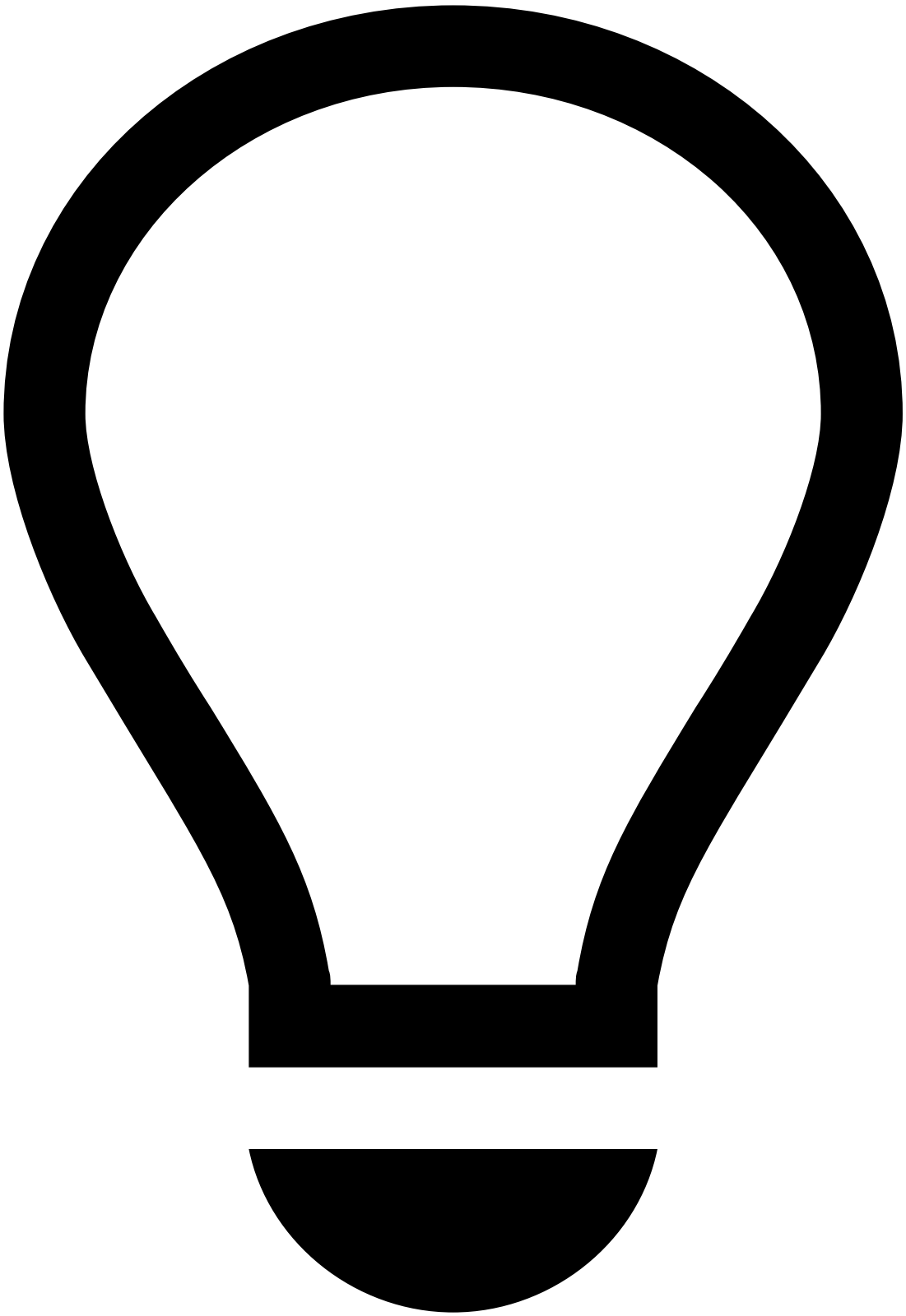
Claim name	Extended name	Description
`iss`	Issuer name	Identifies the issuer of the JWT. This value will be provided by Feedzai, and it should match the tenant id, if applicable.
`exp`	Expiration Time	Date after which the JWT must be rejected.
`nbf`	Not before Time	Date before which the JWT must be rejected.
`jti`	JWT id	Identifier used to revoke JWTs.
`sub`	Subject	Identifies the user of the JWT.

Your final set of claims should be similar to:

```
{
    "iss": "feedzai",
    "exp": 1523022355,
    "nbf": 1523022055,
    "jti": "a00cd6dd-4e55-4694-aeb5-4050e7d063b6",
    "sub": "700e55f9-15b0-4f95-8094-960138a0d886"
}
```



How to generate a JWTâ□□

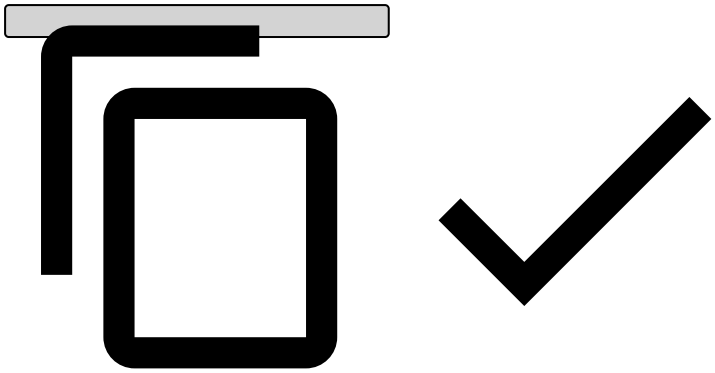


tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

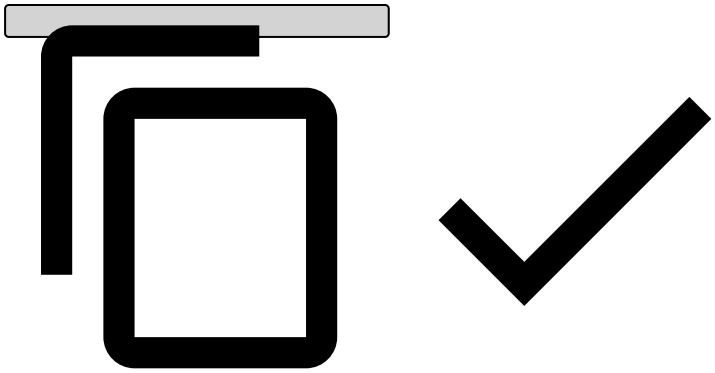
1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```



4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
 --mode JWT \
 --privateKey "/path/to/privateKey" \
 --publicKey "/path/to/publicKey" \
 --iss "issuer" \
 --sub "subject" \
 --exp "$((EPOCHSECONDS + 300))" \
 --kid "key_id"
```



The JWT will be displayed on your standard output. Alternatively, you can use the `--jwt` parameter to specify a file where it should be stored instead.

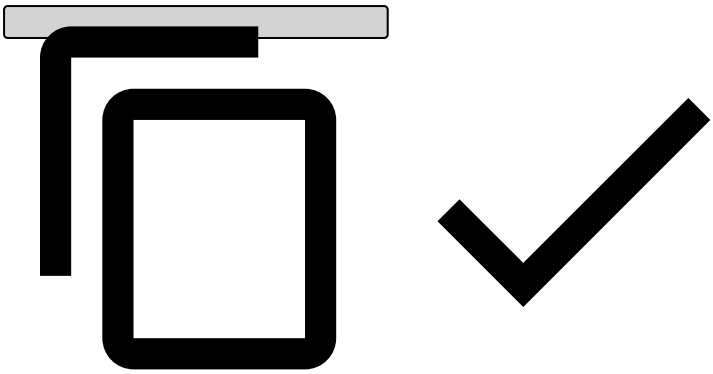
How to send the signed JWTâ

danger

Note that you can only send the signed JWT after the corresponding JWK is uploaded to Pulse (see Manage JWKs in Pulse).

Send the signed JWT in the `Authorization` header of HTTP requests made to Feedzai's API, as follows:

```
Authorization: Bearer <your_jwt>
```



Manage JWKs in Pulseâ□□

Once you create the JWK set, the following JWK steps must be taken:

1. Log into Pulse. Make sure you are accessing the right environment (i.e. **PROD**) and confirm the current list of Authorization keys. Before revoking any key, make sure you are no longer using it.
2. Add the new key to Pulse.
3. Start signing messages with JWTs based on the new key.
4. Revoke the old key.
5. Delete revoked Authorization keys.

List Authorization keysâ□□

To list the existing Authorization keys, access the **Administration** page and select the **Authorization Keys** tab on the sidebar.

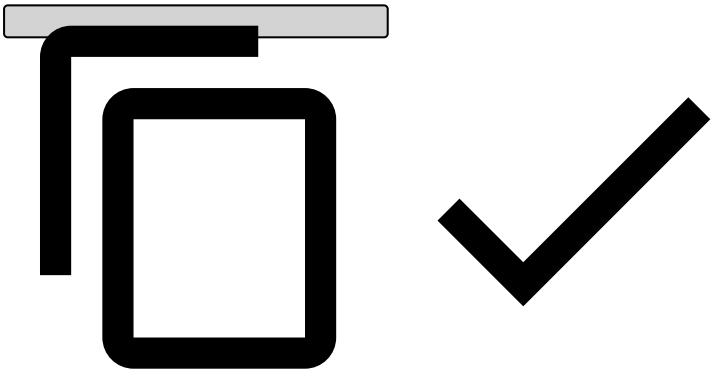
Select a label name or expand to see the Authorization keys associated with that label:

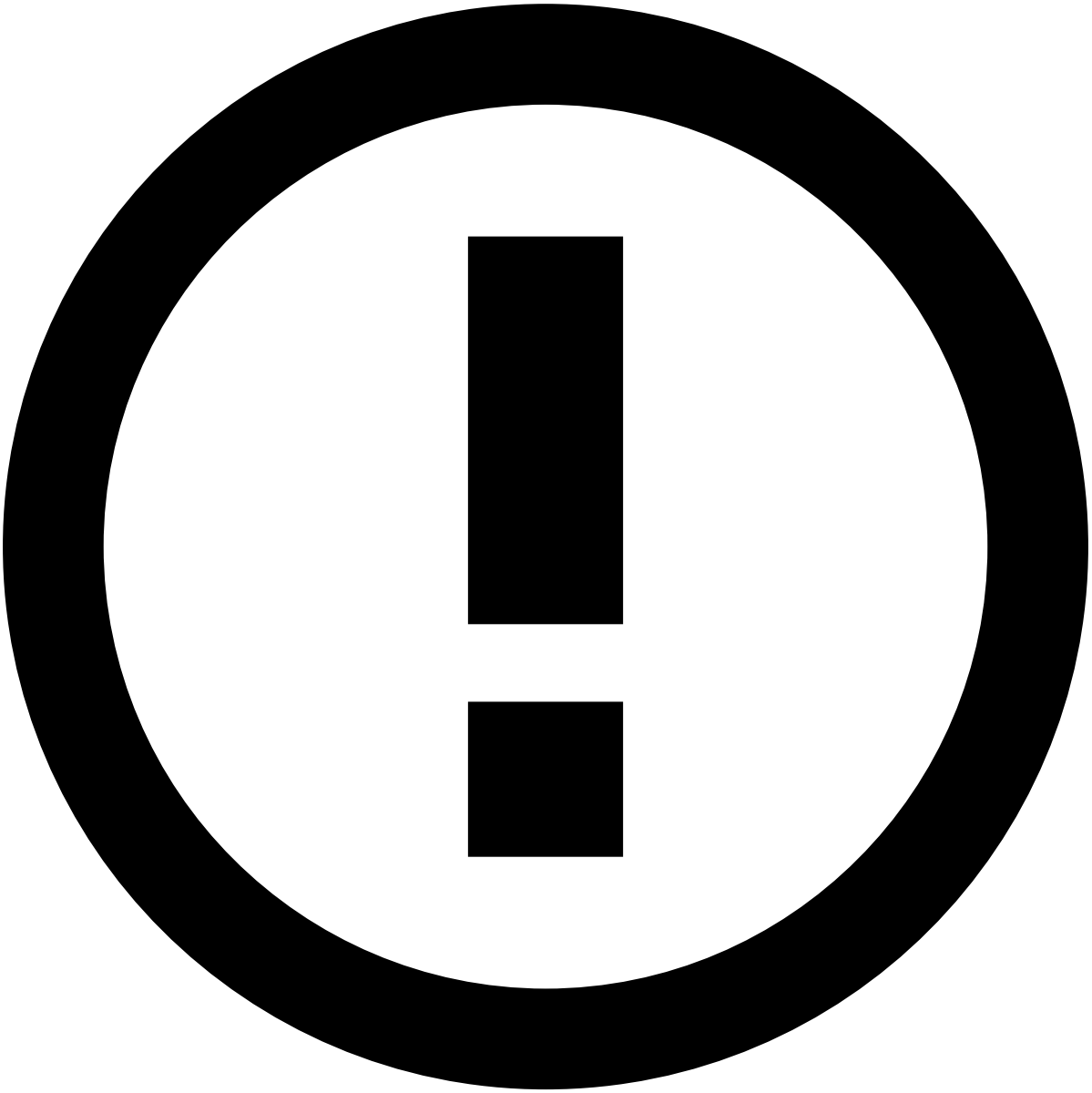
Add Authorization keysâ□□

To add a new Authorization key, select a label name and upload the JSON file that contains the new key:

The JSON file should be an object with the key, using the following format:

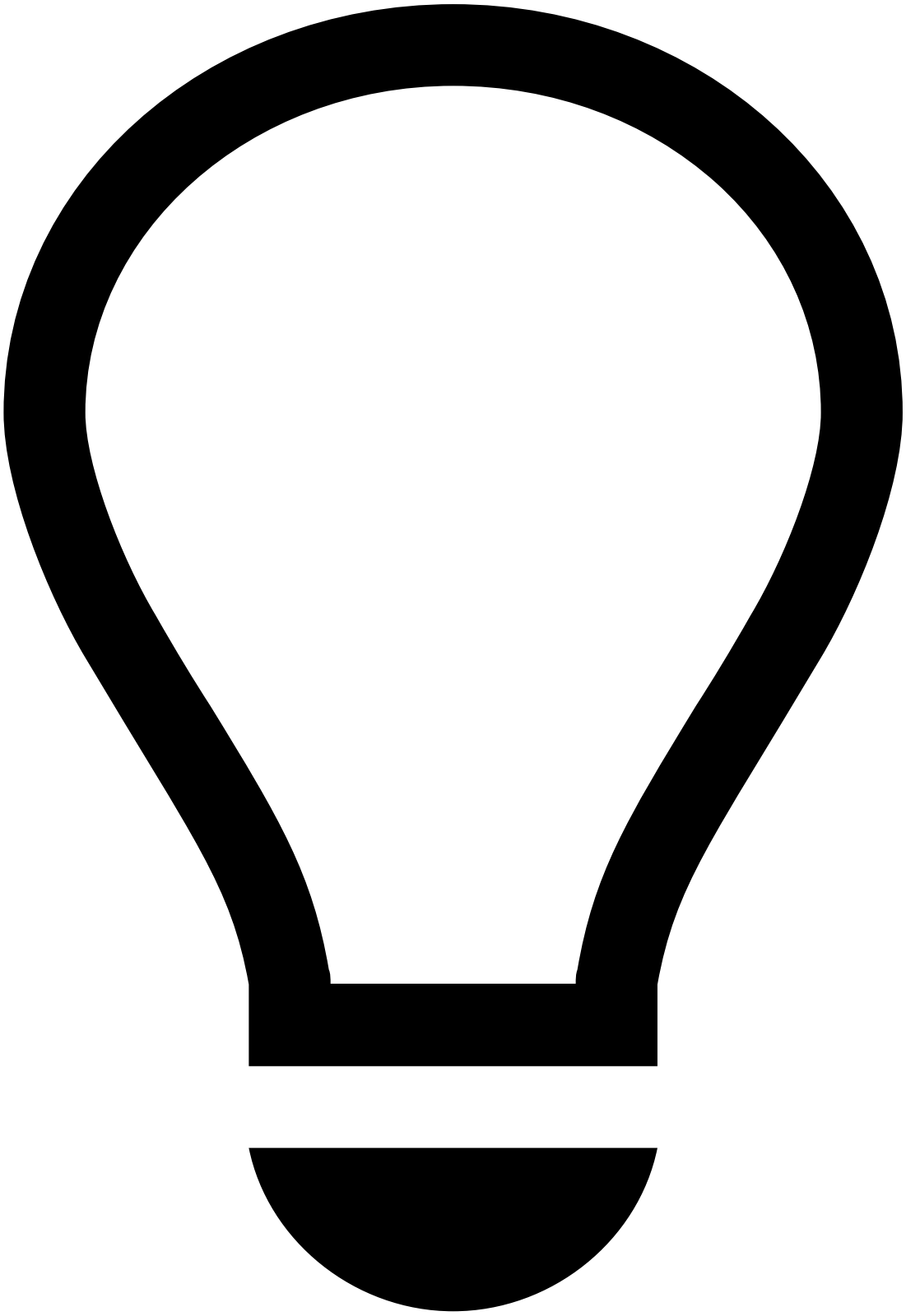
```
{
    "e": "AQAB",
    "n": "187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6
xj9ZnqyyScle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374
XNfLRb0sF7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbO
WgEPUKoRBRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
    "kty": "RSA",
    "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
    "use": "sig"
}
```





info

The new key must be associated with an existing label.



tip

Make sure you start signing the new messages with the new key.

Revoke Authorization keysâ□□

The following animation illustrates how to add a new key and revoke the current one.

danger

An Authorization key should only be revoked once a new Authorization key has been added (see previous step) and JWTs are already being signed with the new key.

Delete revoked Authorization keysâ□□

By deleting an Authorization key, you are permanently removing it from the list of keys. Only revoked keys can be deleted.